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Single-laser-shot femtosecond coherent anti-Stokes Raman scattering thermometry at 1000 Hz in unsteady flames

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Abstract

Temperature measurements at 1000 Hz in driven and turbulent flames are performed using single-laser-shot femtosecond coherent anti-Stokes Raman scattering (fs-CARS). A Hencken burner H_2–air flame is driven at 10 Hz by means of a piston actuator in the air supply line. Temperature measurements are performed by fitting single-laser-shot fs-CARS spectra of N_2. The probe pulse is chirped to map the temporal shape of the Raman coherence into the frequency domain of the CARS signal. The probe pulse is delayed by approximately 2 picoseconds (ps) compared to the simultaneous arrival of the pump and Stokes pulses at the probe volume. The method for calculating theoretical fs-CARS spectra is presented and the procedure for fitting theoretical to experimental spectra is explained. Previous fs-CARS measurements were performed in a heated gas cell and in laminar adiabatic flames. Measurements in a driven H_2–air flame and a turbulent methane–air flame are discussed in this paper. fs-CARS spectra are acquired with excellent signal-to-noise ratios. Temperature excursions in the driven flame due to the movement of the piston actuator in the air line are clearly evident, as well as the fluctuations in the turbulent flame.
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1. Introduction

Coherent anti-Stokes Raman scattering (CARS) has been used for temperature and species measurements in reacting flows for several decades [1]. The introduction of femtosecond (fs) lasers with high repetition rates and high pulse energies paves the way for single-laser-shot CARS measurements performed thousands of times a second. fs-CARS has the potential to temporally resolve turbulent fluctuations. This work will focus on temperature measurements acquired at 1000 Hz in H_2–air flames with a controllable transient phenomenon created by a piston actuator. Preliminary results from a turbulent Bunsen burner methane–air flame are also presented.

fs-CARS experiments have been performed for flame temperature and species concentrations measurements, and have been based on the decay

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of the Raman coherence after initial excitation by the pump and Stokes pulses [2–7]. The temporal decay of the Raman coherence is usually detected by delaying the arrival of the probe pulse relative to the pump and Stokes pulses. The Raman coherence decays because the individual Raman transitions oscillate at slightly different frequencies, and no longer interfere constructively [8,9]. Temperature measurements can be performed because the number of Raman transitions with significant population increases with temperature, and hence the rate at which the initial giant Raman coherence decays will increase at higher temperatures. Species concentration measurements can be performed because different molecules have different Raman transition frequencies and the beating between the Raman transitions can be detected. The decay of the Raman coherence has been monitored from several picoseconds (ps) to hundreds of ps after initial excitation. Time-resolved fs CARS is not a single shot technique and is therefore not capable of resolving turbulent phenomena of interest.

Lang and Motzkus [10] introduced a chirped probe pulse to convert fs-CARS to a single-laser-shot thermometry technique. The chirped probe pulse is typically a few ps in duration and the complete initial decay of the Raman coherence is probed in a single pulse. In a chirped pulse, the characteristic frequency of the pulse varies throughout the pulse. Typically the red wavelengths arrive first and the blue wavelengths last. The chirped probe pulse therefore, maps the temporal decay of the Raman coherence onto the spectrum of the CARS signal pulse. Chirped pulses have also been used to obtain good species selectivity [11]. Roy et al. recently demonstrated chirped probe fs-CARS temperature measurements in near-adiabatic laminar flames [12].

In addition to the high repetition rate of fs lasers used in fs-CARS, this technique has other advantages. Femtosecond lasers have minimal shot-to-shot fluctuations, which aids in modeling the experimental results, and allows for single-laser-shot measurements. By probing only the initial decay of the Raman coherence, collisions can be neglected. Also, it has been demonstrated that the initial decay rate of the Raman coherence in gas phase measurements depends only on temperature [6].

2. Experimental systems

2.1. Driven and turbulent flame systems

Measurements were performed in a driven flame, with perturbations introduced at a known frequency. A 25-mm-square Hencken burner is used to stabilize a nonpremixed laminar H_2–air flame with a N_2 guard flow surrounding the flame. Digital mass flow controllers are calibrated and used to control the volumetric flow rates of the H_2, air, and N_2. To create a controlled transient event in the flame, a piston actuator device is installed in the air supply line, as shown in Fig. 1. The mass flow controller is located before the piston device and the set point on the mass flow controller is kept constant. A function generator is used to trigger the piston device at 10 Hz. When the piston is actuated, extra air is forced into the air line, and the overall flame equivalence ratio momentarily decreases. The piston then retracts causing a lower air flow rate, briefly increasing the overall equivalence ratio of the flame. Similar devices have been used to study fundamental flame–vortex interactions experimentally [13] and numerically [14]. Measurements were also performed in a turbulent methane–air Bunsen burner flame. While the flow rates of the reactants and characteristic frequencies of this flame are not well characterized, it provides an environment with random temperature fluctuations. The amount of turbulence could be controlled by adjusting the fuel flow rate and the amount of premixed air.

2.2. fs-CARS system

A Ti:sapphire laser with regenerative amplification emits 1 mJ, 800 nm, 85 fs full-width-half-maximum (FWHM) pulses at a repetition rate of 1000 Hz. This laser is used to pump an optical parametric amplifier (OPA) and also to provide the Stokes pulse with an approximate energy of 100 μJ/pulse. The frequency-doubled output of the OPA, centered at 675 nm with an energy of 30 μJ/pulse, is used for both the pump and probe beams. A 50% beam splitter is used and the reflected portion is used as the probe pulse. The probe pulse is chirped by transmission through a 30 cm long glass rod, causing the temporal width of the pulse to increase from about 85 fs to about 2.5 ps FWHM. A folded-BOXCARS geometry is utilized with the pump and Stokes pulses in the same horizontal plane. The CARS signal, centered at approximately 584 nm,

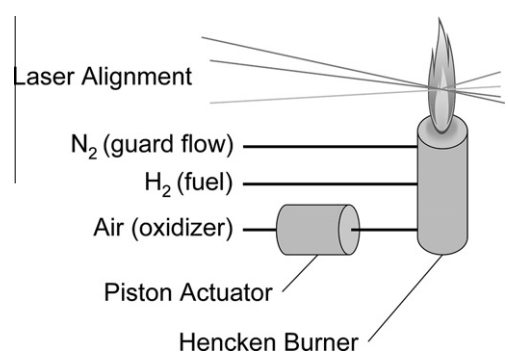

Fig. 1. Schematic diagram of driven flame apparatus.

is then filtered using a short-pass filter before being detected with a 0.25 m spectrometer and an electron-multiplying charge-coupled-device (EMCCD) camera. Atomic emission lamps are used to calibrate the detection system, and the spectral resolution is found to be 0.042 nm/pixel. The principal features in the fs-CARS spectra cover at least 10 pixels and no instrument function convolution is needed when comparing experiment to theory. Computer controlled mechanical translation stages control the time of arrival of the three pulses at the CARS probe volume location. The pump and Stokes pulses are overlapped in time by maximizing the sum frequency generation signal in a thin nonlinear crystal at the focus plane. The spatial overlap of the three beams is optimized by attenuating the pulses and inserting a pinhole aperture at the focus plane. A 300 mm focal length lens is used to focus the beams to the probe volume, and it is estimated that the probe volume length is 1 mm.

In order to acquire single-laser-shot spectra at 1000 Hz, the EMCCD camera is operated in a full vertical binning, cropped mode, using an area 30×800 pixels. Operating the EM mode of the camera decreases the speed, and sufficient signal levels are maintained to avoid using this functionality of the camera in all data presented here. The camera could be made to record spectra at higher repetition rates by using a smaller cropped area. Horizontal binning would also increase the speed of the camera, but would sacrifice spectral resolution. The fs-CARS spectra acquired in a gas cell (presented below) was recorded using a horizontal bin size of two, and sufficient spectral resolution is maintained to compare with the theoretical model. The authors have also done preliminary work using an image intensifier and complimentary metal–oxide–semiconductor (CMOS) camera as the detection system. This system can record single-laser-shot spectra at data rates of at least 10 kHz, but introduces additional noise.

3. Theoretical model

As is the case with ns-CARS, the temperature is determined from the experimental spectra by finding the best-fit theoretical spectrum with temperature as one of the fitting variables. The theoretical model for the chirped-probe-pulse (CPP) fs-CARS spectrum has some significant differences compared to the model for ns-CARS and is briefly outlined here. The CPP fs-CARS measurement method is illustrated schematically in Fig. 2. It is first assumed that the pump and Stokes pulses excite the vibrational Raman transitions of the N_2 molecule with the same phase. Furthermore, it is assumed that the magnitude of the Raman polarization for each transition is proportional to the Raman cross-section and to the population difference between the lower and upper levels of each transition. These assumptions are supported by detailed analysis of the Raman

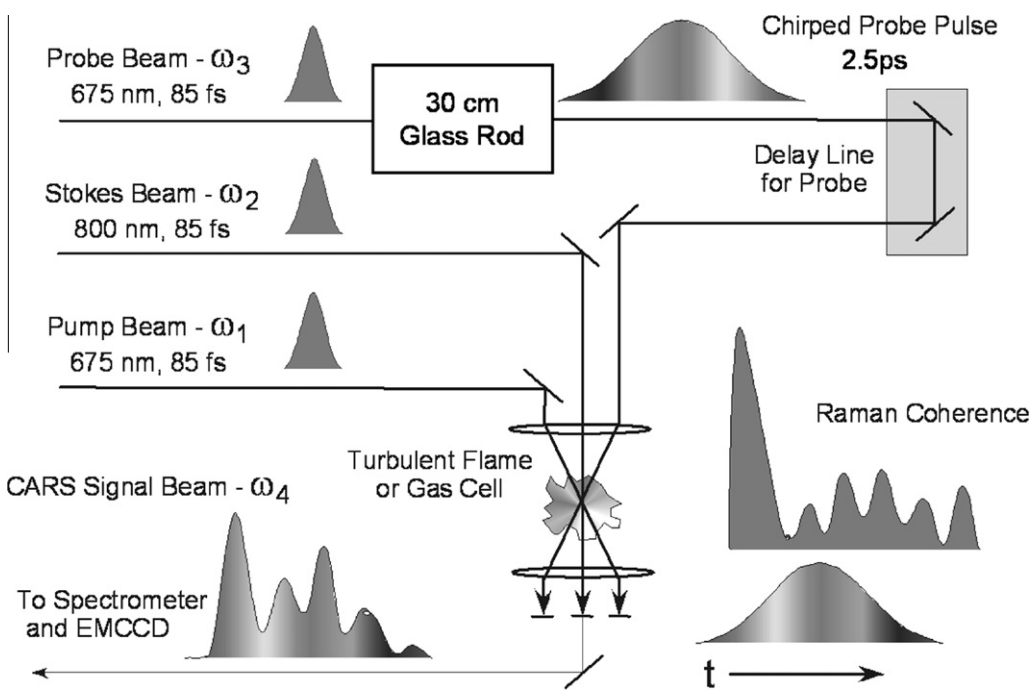

Fig. 2. Schematic diagram of optical system for the chirped probe pulse fs-CARS experiment.

excitation process based on numerical integration of the time-dependent density matrix equations [8]. Therefore the Raman polarization induced by the pump and Stokes beams is given by

$$P_{res}(t) = \beta\left[\int_{-\infty}^{t} E_p(t')E_S^*(t')dt'\right] \times \sum_i \left\{\Delta N_i \left(\frac{d\sigma}{d\Omega}\right)_i \cos(\omega_i t + \phi)\exp(-\Gamma_i t)\right\} \quad (1)$$

In Eq. (1), $E_p(t)$ and $E_S(t)$ are the electric-field amplitudes of the pump and Stokes pulses, respectively, ΔN_i and $(d\sigma/d\Omega)_i$ are the population difference and Raman cross-section for each Raman transition, i, respectively, and β is the resonant intensity scaling factor. The Stokes and pump electric-fields are modeled as transform limited Gaussian pulses, and the pump and Stokes temporal pulse widths are fitting parameters. The population distribution is temperature dependent, and the terms ΔN_i are calculated within the fitting routine as the temperature is a fitting parameter. The phase factor, ϕ, is used to account for the different phase of the resonant and nonresonant polarizations and is a fitting parameter. Note that at time $t = 0$ the phase of each transition is assumed to be the same; in other words, the Raman transitions oscillate in phase at time $t = 0$. As can be seen in Eq. (1), the resonant polarization (Raman coherence) grows proportionally to the integrated value of the product of the Stokes and pump electric-fields. After excitation by the pump and Stokes beams, the polarization for the various Raman transitions oscillates with angular frequency ω_i, and the total resonant polarization experiences destructive interference. The polarization for individual transitions decays as a result of dephasing collisions with a rate constant of Γ_i, the Raman linewidth. However, for the time scales of a few picoseconds after the impulsive pump–Stokes Raman excitation, $\exp(-\Gamma_i t) = 1$ to a very good approximation. The nonresonant polarization is given by

$$P_{nres}(t) = \alpha E_p(t)E_S(t) \quad (2)$$

In Eq. (2), α is the nonresonant scaling parameter.

The CARS and the nonresonant four-wave mixing signal result from the scattering of the probe beam from the induced Raman polarization and nonresonant polarization, respectively. The probe beam is assumed to be a linearly chirped pulse, and the electric-field is given by

$$E_{pr}(t) = E_{0,pr} \times \exp\left[-(t - t_{0,pr})^2/\tau_{pr}^2\right]\cos\left(\omega_{0,pr}t - \beta_{pr}t^2\right) \quad (3)$$

The probe pulse width τ_{pr}, chirp parameter β_{pr}, and probe time delay $t_{0,pr}$ are fitting parameters. Note that the probe field is a real quantity, as is also the case with the pump and Stokes fields. The time dependence of the electric-field $E_{sig}(t)$ of the signal is given by

$$E_{sig}(t) = E_{pr}(t)P_{res}(t) + E_{pr}(t)P_{nres}(t) \quad (4)$$

The spectral intensity of the signal is found by taking the Fourier transform of the time-dependent electric-field and then squaring the magnitude:

$$E_{sig}(\omega) = \int_{-\infty}^{+\infty} [E_{pr}(t)P_{res}(t) + E_{pr}(t)P_{nres}(t)] \times \exp(i\omega t)dt \quad (5)$$

$$S_{CARS}(\omega) = |E_{sig}(\omega)|^2 \quad (6)$$

The CARS spectral intensity $S_{CARS}(\omega)$ is measured experimentally using the spectrometer and EMC CD camera. Recently, we have begun to incorporate higher order nonlinear chirp for the probe beam and linear chirp in the pump and Stokes beams.

4. Results and discussion

To test the accuracy and precision of CPP fs-CARS, measurements were first performed in a heated gas cell filled with air at atmospheric pressure. The gas cell was held at 300, 500, 700 and 900 K while single-laser-shot spectra were recorded. The accuracy of the gas cell temperature controller is estimated to be ±2 K. Figure 3a shows a single-laser-shot N_2 CARS spectrum acquired from the gas cell at 700 K, and the corresponding result of the fitting code ($T = 681$ K). Figure 3b shows a histogram of best fit temperatures from 1000 consecutive single-laser-shot spectra at 700 K. For the data in Fig. 3, the probe time delay was 2 ps.

To obtain values for many of the fitting parameters, the following procedure was used. The average of many (typically 1000) spectra at a known temperature, either room temperature or an adiabatic laminar flame temperature, was fit while holding the temperature fixed at the known temperature. The following parameters were varied in this initial room temperature fit: the pump, Stokes, and probe pulse widths, the chirp parameter for the probe pulse, the probe time delay, the relative phase between the resonant and nonresonant polarizations, and the resonant and nonresonant scaling parameters. The values for the pulse width parameters and the probe time delay were typically close to the expected values. The value for the relative phase between the resonant and nonresonant polarizations consistently converged to a value between 255° and 275°. The ratio of the scaling parameters (β/α) varies primarily with probe time delay, and secondarily with temperature. At room temperature the value was 3.8e8 for a probe time delay of 2 ps, and for flame

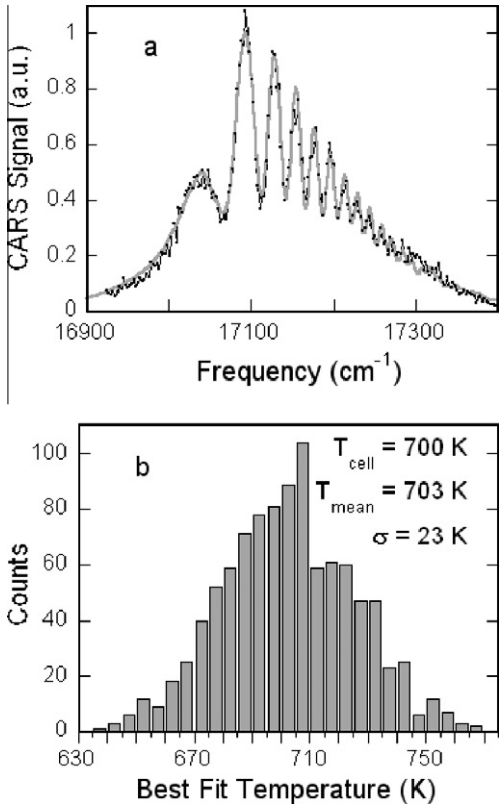

Fig. 3. fs-CARS temperature measurements made in a heated gas cell at 700 K. (a) A single-laser-shot spectrum (black symbols and line) and the best fit calculated spectrum (grey line). The temperature of the best fit spectrum is 681 K. (b) A histogram of the best fit temperatures of 1000 consecutive single-laser-shot spectra.

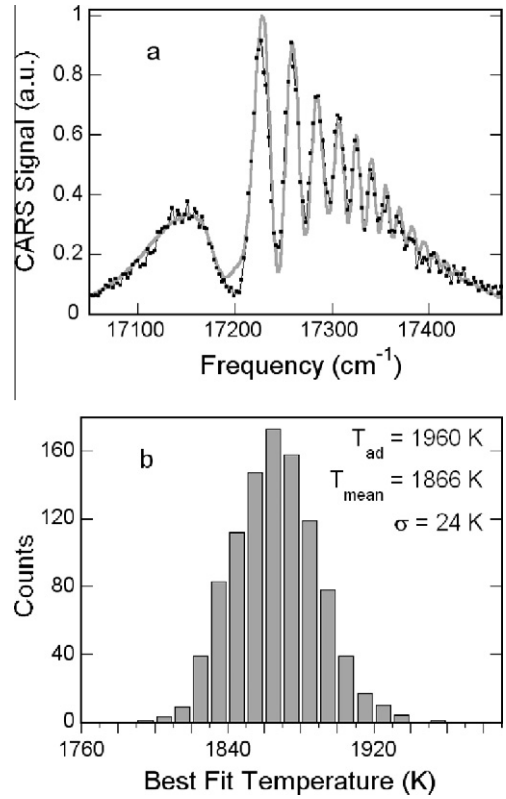

Fig. 4. fs-CARS temperature measurements made in a laminar steady-state Hencken burner flame with equivalence ratio of 0.7. (a) A single-laser-shot spectrum (black symbols and line) and the best fit calculated spectrum (grey line). The temperature of the best fit spectrum is 1887 K. (b) A histogram of the best fit temperatures of 1000 consecutive single-laser-shot spectra.

equivalence ratios from 0.3 to 0.7 the value was close to 1.5*e*8 for a probe time delay of 1.5 ps.

Spectra acquired from the gas cell and flames were fit with these parameters fixed at the values determined from fitting the room temperature data, while varying temperature. The average best fit temperature for the 1000 single-laser-shot spectra acquired from the gas cell at the fixed temperature of 700 K is 703 K and the standard deviation is 23 K, or about 3% of the mean temperature.

Temperature measurements were also performed in steady-state near-adiabatic laminar H_2–air flames on the same Hencken burner described above [12]. A single-laser-shot spectrum acquired from a flame at an equivalence ratio of 0.7 is shown in Fig. 4a and b shows the histograms of best fit temperatures from 1000 consecutive single-laser-shot spectra in this flame. This data was acquired with a probe time delay of 1.5 ps, and the laser parameters were found by fitting an average flame spectrum. For adiabatic flames with equivalence ratios from 0.5 to 1.0, fs-CARS exhibited excellent precision. For flames with equivalence ratios of 0.3, 0.5, 0.7, and 1.0, the standard deviation of 1000 consecutive single-laser-shot temperature measurements divided by the mean temperature was 1–1.5%. The mean temperatures determined from the fs-CARS measurements were within 6% of the calculated adiabatic flame temperatures. The accuracy of the CPP fs-CARS technique is at this point limited by our lack of characterization and monitoring equipment for the laser pulses, and should improve markedly as we start to use equipment to measure the phase and spectrum of all three laser pulses. We also must develop a better understanding of the effect of the OPA on the spectrum and phase of the pump and probe pulses.

Temperature measurements were also performed in the driven flame system described above. The unperturbed flame equivalence ratio is 0.5 and the probe time delay was 1.5 ps. The measured temperature accurately captures

the 10 Hz periodic fluctuations introduced by the piston actuator, as shown in Fig. 5. As expected, the measured flame temperature decreases as the piston is actuated and the flame becomes momentarily more lean. The measured flame temperature then increases above the steady-state temperature as the piston retracts, causing a momentary increase in the equivalence ratio. Figure 5 also shows the time history of best fit temperatures from a laminar steady-state flame at the same equivalence ratio, but at a slightly different flame position, for comparison with the driven flame. Figure 6 shows the average and standard deviation of the measured temperature for 10 piston actuations. From this figure, it is evident that the precision of the measurement does not vary significantly as the flame temperature varies over almost 200 K. Four single-shot spectra and the best-fit CPP fs-CARS spectra acquired during the temperature excursion between 360 and 380 ms are shown in Fig. 7. The spectra from

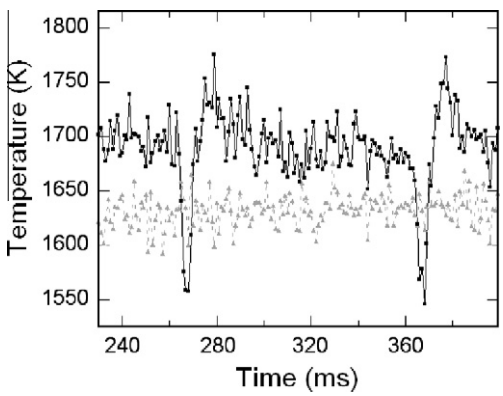

Fig. 5. fs-CARS temperature measurements made in a driven H_2–air flame (black solid line and symbols) and a steady-state flame (grey dashed line and symbols).

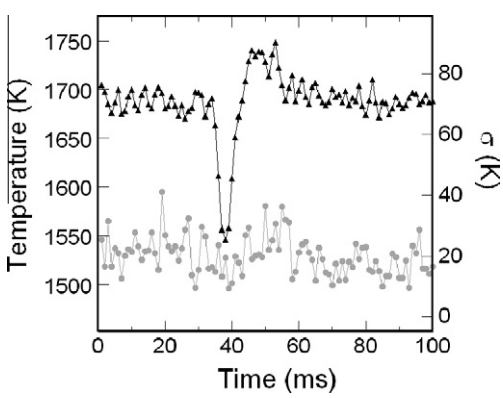

Fig. 6. The average (black) and standard deviation (grey) of the best fit temperature for 10 piston actuations in the driven flame described in the text.

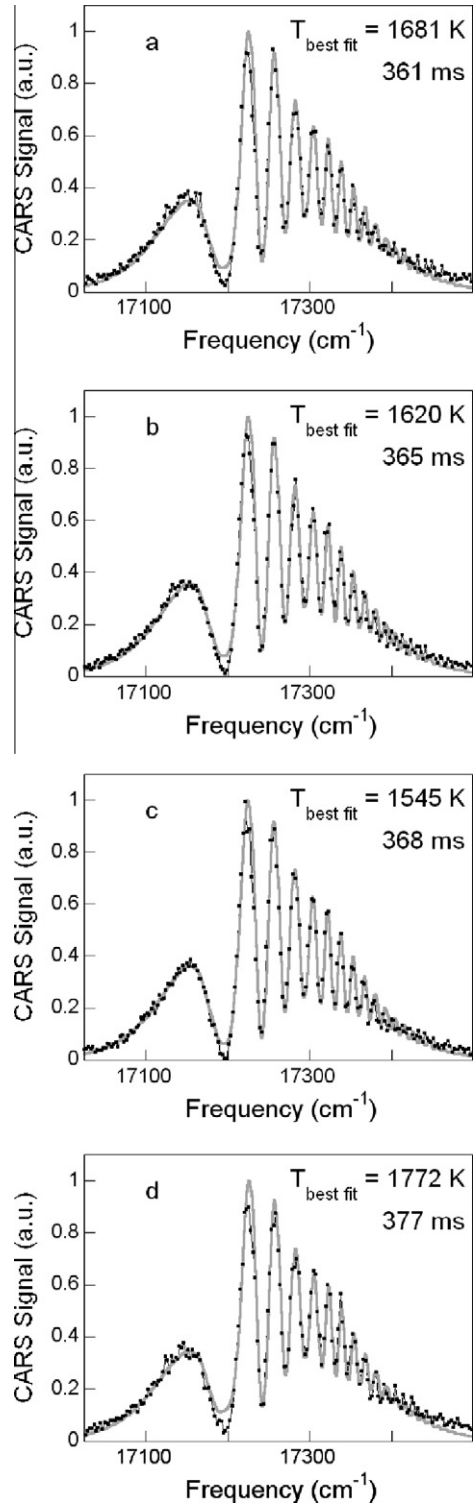

Fig. 7. Single-laser-shot fs-CARS spectra (black symbols and line) at four times of interest in the driven flame, with the corresponding best fit calculated spectra (grey line).

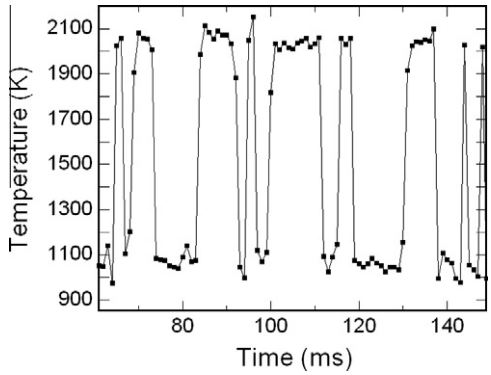

Fig. 8. A time history of the best fit temperature from single-laser-shot spectra taken in a turbulent methane–air flame.

the driven flame exhibit excellent signal-to-noise characteristics.

fs-CARS spectra were also recorded in the turbulent methane–air flame describe above. The probe time delay was 1.0 ps. Fluctuations are clearly evident in the time history of best fit temperatures shown in Fig. 8. The time history and histogram (not shown) of best fit temperatures reveal a bimodal flame structure, which is probably a result of the flame tip moving into and out of the probe volume location. The measurements were performed approximately 8 cm above the exit of the burner. The single-shot fits for this data are comparable to those shown in Fig. 6.

5. Conclusion

Temperature measurements in driven and turbulent flames have been performed using single-laser-shot CPP fs-CARS spectra of N_2. The repetition rate of the laser used in this study is 1 kHz and transient features in the flames were accurately captured. The authors have ordered a 10 kHz laser system with similar pulse energies and anticipate no new challenges in recording fs-CARS spectra at this data rate. Some modifications to the detection system will need to be made. The theory of CPP fs-CARS and the fitting routine for determining temperatures from the experimental spectra were discussed. The demonstrated precision of the CPP fs-CARS techniques for flame measurements is excellent, with measured temperature standard deviations of less than 2% of the mean temperature over a wide range of flame conditions. The accuracy of the technique is on the order of 5% of the mean temperature for the measurements reported here, but the accuracy of the measurements should improve markedly in the future when equipment to monitor the spectrum and phase of the laser pulses is used and this information is incorporated in the theoretical analysis of the spectra. This work highlights the potential use of CPP fs-CARS for performing temperature measurements in turbulent flames at data rates on the order of a few kHz.

Acknowledgments

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